



Welcome to AWS Academy Data Engineering

Module overview

Sections

- Course prerequisites and objectives
- Course overview

Course prerequisites and objectives

Welcome to AWS Academy Data Engineering

Prerequisites

- Completed the AWS Academy Cloud Foundations course or have equivalent experience
- Worked with Structured Query Language (SQL)
- Worked with databases
- Introduced to general networking concepts
- Understanding of decision-making knowledge in math, probability, and statistics

Related job roles

- Data engineer
- Data analyst
- Data scientist
- Extract, transform, and load (ETL) developer
- Machine learning (ML) practitioners

Course objectives (Slide 1 of 2)

This course prepares you to do the following:

- Summarize the role and value of data science in a data-driven organization.
- Recognize how the elements of data influence decisions about the infrastructure of a data pipeline.
- Illustrate a data pipeline by using AWS services to meet a generalized use case.
- Identify the risks and approaches to secure and govern data at each step and each transition of the data pipeline.
- Identify scaling considerations and best practices for building pipelines that handle large-scale datasets.
- Design and build a data collection process while considering constraints such as scalability, cost, fault tolerance, and latency.

Course objectives (Slide 2 of 2)

- Select a data storage option that matches the requirements and constraints of a given data analytics use case.
- Implement the steps to process both structured, semistructured, and unstructured data formats in a data pipeline that is built with AWS.
- Explain the concept of MapReduce and how Amazon EMR is used in big data pipelines.
- Differentiate the characteristics of an ML pipeline and its specific processing steps.
- Analyze data by using AWS tools that are appropriate to a given use case.
- Implement a data visualization solution that is aligned to an audience and data type.

Course overview

Welcome to AWS Academy Data Engineering

Course outline

- Module 1: Welcome to AWS Academy Data Engineering
- Module 2: Data-Driven Organizations
- Module 3: The Elements of Data
- Module 4: Design Principles and Patterns for Data Pipelines
- Module 5: Securing and Scaling the Data Pipeline
- Module 6: Ingesting and Preparing Data
- Module 7: Ingesting by Batch or by Stream
- Module 8: Storing and Organizing Data
- Module 9: Processing Big Data
- Module 10: Processing Data for ML
- Module 11: Analyzing and Visualizing Data
- Module 12: Automating the Pipeline
- Module 13: Bridging to Certification
- IoT Use Case Slides

Course modalities (Slide 1 of 2)

Slides

- All modules

Activities

- Module 3: Planning Your Pipeline
- Module 4: Using the Well-Architected Framework
- Module 10: Labeling with SageMaker Ground Truth

Demonstrations

- Module 10:
 - Preparing Data and Training a Model with SageMaker
 - Preparing Data and Training a Model with SageMaker Canvas
 - Creating a Linear Regression Model Using Amazon Q Developer

Course modalities (Slide 2 of 2)

Labs

- Module 4: Querying Data by Using Athena
- Module 7: Performing ETL on a Dataset by Using AWS Glue
- Module 8: Storing and Analyzing Data by Using Amazon Redshift
- Module 9:
 - Processing Logs by Using Amazon EMR
 - Updating Dynamic Data in Place
- Module 11: Analyzing and Visualizing Streaming Data with Amazon Data Firehose, OpenSearch Service, and OpenSearch Dashboards
- Module 12: Building and Orchestrating ETL Pipelines by Using Athena and Step Functions

Capstone Project

Learning assessments

Knowledge checks

- Provided for modules 2 through 12
- Drawn from the slides and student guide notes

Course assessment

- 25 randomized questions
- Drawn from the slides and student guide notes

Preparing for certification

- This course is not intended to fully prepare you for related AWS Certifications.
- Review the Exam Guides for information regarding the certifications.
- The Bridging to Certification module provides information about additional exam preparation resources.

Thank you



Corrections, feedback, or other questions?

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